Course Name: Computer Networks and Internet Protocol

Assignment - Week 5 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1

Sender Window in TCP can be given as:

a) Sender Window = max(Congestion Window, Receiver Window)
b) Sender Window = Congestion Window
c) Sender Window = Receiver Window
d) Sender Window = min(Congestion Window, Receiver Window)

Correct Answer: (d)
Detailed Solution: The Sender Window is computed as the minimum of the Congestion Window and the Receiver Window.

QUESTION 2

What is the advantage of an iterative server over a concurrent server?

a) It handles multiple requests faster.
b) It is easier to implement and debug.
c) It supports asynchronous I/O.
d) It has lower response times for high-traffic scenarios.

Correct Answer: (b)
Detailed Solution: An iterative server is simpler to implement and debug due to its sequential nature, making it easier to manage than concurrent server models.

QUESTION 3

Which of the following uses UDP?
a) DNS
b) POP
c) HTTP
d) FTP

Correct Answer: (a)

Detailed solution: DNS primarily uses UDP on port 53 for speed and efficiency. Because DNS queries are generally small and fit within a single packet, UDP avoids the overhead of establishing a connection (no 3-way handshake), allowing for rapid, low-latency lookups. It is ideal for high-volume, quick transactions.

QUESTION 4

What impact does a higher α value have on Jacobson's algorithm?

a) It makes SRTT more sensitive to sudden RTT changes.
b) It reduces the impact of new RTT measurements on SRTT.
c) It increases the weight of RTTVAR in the RTO calculation.
d) It eliminates the need for RTTVAR.

Correct Answer: (b)

Detailed solution: In Jacobson's algorithm, α\alphaα controls the smoothing of SRTT. A higher α\alphaα places more emphasis on historical RTT values, reducing the influence of new RTT measurements. This leads to a slower adaptation to sudden RTT changes, which may provide more stability but less responsiveness. Conversely, a lower α\alphaα increases sensitivity to sudden RTT variations but can make SRTT less stable.

QUESTION 5

FD_ZERO is used for

a) tests to see if a file descriptor is part of the set
b) Initializes the file descriptor set FD_SET – a bitmap of fixed size
c) Set a file descriptor as a part of an fd_set
d) None of the above

Correct Answer: (b)

Detailed solution: FD_ZERO initializes an fd_set by clearing all bits, so the set starts empty.
It prepares the descriptor set before adding any file descriptors with FD_SET.

QUESTION 6

SOCK_DGRAM represents

a) UDP-based Datagram Socket
b) TCP-based Stream Socket
c) QUIC-based Stream Socket
d) None of the above

Correct Answer: (a)

Detailed Solution: SOCK_DGRAM represents a UDP-based Stream Socket.

QUESTION 7

select() system call returns one when?

a) Exactly one monitored file descriptor becomes ready for read/write or reports an error condition.
b) This means the call timed out without any event ready for the sockets monitored
c) Zero is the number of sockets that have events pending (read, write, exception)
d) OS kills the process.

Correct Answer: (a)

Detailed Solution: select() returns 1 when exactly one file descriptor is ready for I/O or has an exceptional (error) condition. It counts that FD as "ready," so the return value becomes 1.

QUESTION 8

What does the listen() function do in socket programming?

a) It binds a socket to an address.
b) It puts the socket into a passive mode to wait for incoming connections.
c) It actively connects to a server socket.
d) It sends data over the socket.

Correct Answer: (b)

Detailed Solution: The listen() function sets up a socket to accept incoming connections by placing it in passive listening mode.

QUESTION 9

Which of the following is false?

a) TCP uses congestion control
b) TCP uses both flow control and congestion control
c) TCP uses flow control
d) TCP doesn’t have any flow control, congestion control

Correct Answer: (d)

Detailed solution: TCP has flow control, congestion control. So option (d) is false.

QUESTION 10

Which function can be used to configure socket options like timeout or buffer size?

a) setsockopt()
b) sockopt()
c) configure()
d) options()

Correct Answer: (a)

Detailed Solution : The setsockopt() function is used to configure various socket options, such as timeout values, buffer sizes, and other protocol-specific settings.
